

Putting Rational and Irrational Numbers in Order

Estimating a square root, comparing two quantities, and ordering a mixed list

Before you start. A list can mix fractions, decimals and square roots. They can only be ordered once they are all written the same way.

- **Trap the root** between the two perfect squares on either side of the radicand, then refine.
- **Convert everything to a decimal** — and carry enough decimal places that no two values still look equal.
- **Two positive numbers can also be compared by squaring** them, since squaring keeps the order.
- On the negative side of zero the order **reverses**: the number furthest from zero is the least.

1. Between which two consecutive whole numbers does $\sqrt{50}$ lie? Use that to check yourself, then write $\sqrt{50}$ rounded to the nearest hundredth.

Answer: _____

2. Write $>$, $<$ or $=$ between the two values.

$\sqrt{9}$ _____ 2.9

Answer: _____

3. Order these from least to greatest. Show the decimal you used for each one.

$$3.5, \quad \sqrt{10}, \quad \frac{10}{3}$$

Answer: _____

4. Write $\sqrt{20}$ rounded to the nearest hundredth, and name the two consecutive whole numbers it lies between.

Answer: _____

5. Write $>$, $<$ or $=$ between the two values. Squaring both is a quick way to be sure.

$$\sqrt{2} \quad \underline{\hspace{1cm}} \quad 1.5$$

Answer: _____

6. Order these from least to greatest.

$$\sqrt{30}, \quad 5.5, \quad \frac{27}{5}$$

Answer: _____

7. Many people are taught that π “is” $\frac{22}{7}$. Write $>$, $<$ or $=$ between them, using enough decimal places to be certain.

$$\pi \quad \text{_____} \quad \frac{22}{7}$$

Answer: _____

8. Write $\sqrt{85}$ rounded to the nearest hundredth. Say which perfect square it is closer to, and why that tells you the answer is not half-way between the whole numbers.

Answer: _____

- 9.** Order these from *greatest* to *least* — read the direction carefully.

$$\sqrt{45}, \quad 6.7, \quad \frac{20}{3}$$

Answer: _____

- 10.** Two values that a calculator display can make look equal.

(a) Write $\frac{17}{4}$ as a decimal.

(b) Now write $>$, $<$ or $=$ between the two values.

$$\sqrt{18} \quad \text{_____} \quad \frac{17}{4}$$

Answer: _____

11. Order these four values from least to greatest. Two decimal places will not separate all of them — carry more.

$$\sqrt{2}, \quad 1.42, \quad \frac{7}{5}, \quad 1.41$$

Answer: _____

12. Synthesis. Order these five values from least to greatest, then sketch a number line and mark each one on it.

$$-\sqrt{5}, \quad -2.3, \quad -\frac{9}{4}, \quad \sqrt{6}, \quad 2.4$$

Answer: _____